

GNN Verification with GraphStar Sets

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Anne Tumlin

Vanderbilt University

Why Verify Neural Networks?

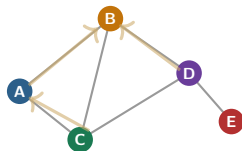
Testing $\neq$ Guarantees

- Neural networks deployed in **safety-critical systems**: power grids, autonomous vehicles, molecular design
- Testing covers individual inputs — but what about **all possible inputs** in a region?
- **Verification**: mathematically prove that a property holds for ALL inputs within a perturbation bound
- Key question: if the input is perturbed by ε , can the output change class?

What is a Graph Neural Network?

Graphs as structured input

- Input: a graph $G = (V, E)$ with node features X and (optionally) edge features E
- Unlike feedforward NNs: the **topology** matters
- **Message passing**: each node collects information from its neighbors
- After k layers, each node's representation captures its k -hop neighborhood
- Applications: molecular property prediction, power flow, social networks

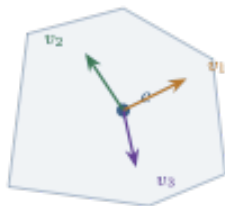


Message passing along edges

Verification with GraphStar Sets

The Shape of Uncertainty

Star Set
(Feedforward NN Input)



Each generator v_i is a vector

$$v_1 = \begin{bmatrix} 0.3 \\ 1.2 \\ -0.4 \\ 0.8 \end{bmatrix} \quad \begin{matrix} \updownarrow \\ n \end{matrix}$$

GraphStar Set
(GNN Input)



Each generator V_i is an $N \times F$ matrix

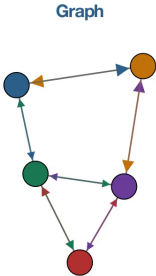
$$V_1 = \begin{matrix} & f_1 & f_2 & f_3 & f_4 \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0.2 & -0.1 & 0.5 & -0.3 \\ 0.2 & -0.1 & 0.5 & -0.3 \\ 0.2 & -0.1 & 0.5 & -0.3 \\ 0.2 & -0.1 & 0.5 & -0.3 \\ 0.2 & -0.1 & 0.5 & -0.3 \end{bmatrix} & \begin{matrix} \updownarrow \\ N \end{matrix} \end{matrix}$$

each row = a node

GNN Message Passing

How information flows through a GNN layer

GNN Message Passing: Graph & Matrix Representation



Node Feature Matrix X

	f_1	f_2	f_3
A	1.0	0.5	-0.2
B	0.3	-0.8	1.2
C	-0.5	0.7	0.4
D	0.8	0.1	-0.6
E	-0.3	0.9	0.2

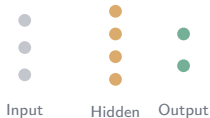
All nodes aggregate simultaneously

Step 1: Aggregate — each node gathers neighbor features

Perturbation Propagation: FFNN vs GNN

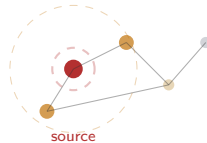
Feedforward NN

Perturbation mixes across **features** through weight matrices



Graph Neural Network

Perturbation spreads across **topology** hop-by-hop



See animated version: `fig2_perturbation_propagation_v2.gif`

GNN Verification: Set Propagation

How a GraphStar flows through one GCN layer

1. **Aggregate**: Multiply each basis matrix by $\hat{A}$ (all at once)
Constraints pass through unchanged (linear operation)
2. **Transform**: Multiply by weights W , add bias b
Still linear — constraints still unchanged
3. **Flatten**: Reshape GraphStar $(N \times F \times p) \rightarrow \text{Star } (NF \times p)$
Reuses all existing NNV Star-based ReLU machinery
4. **ReLU reachability**: LP solves for uncertain neurons ($+k$ predicates)
This is where verification complexity lives
5. **Reshape**: Star back to GraphStar (restore graph structure)

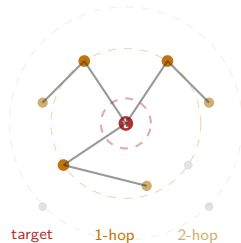
$$\text{GraphStar} \xrightarrow{\hat{A} \cdot V \cdot W + b} \text{flatten} \xrightarrow{\text{ReLU reach.}} \text{reshape} \rightarrow \text{GraphStar}'$$

Subgraph Verification: Exact & Scalable

- A k -layer GNN's output for node v depends **only** on its k -hop neighborhood
- Extract the subgraph, verify just that piece
- **Exact** result — not an approximation
- 118-node graph $\rightarrow$ verify ~ 20 -node subgraphs

See animated version:

`fig4_subgraph_verification_v2.gif`



Node + Edge Perturbation

What does it mean to perturb node features vs. edge features?

Node Perturbation (δ)



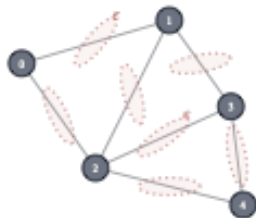
Each node's features can vary:

$$x_i \in [\bar{x}_i - \delta, \bar{x}_i + \delta]$$

"How much can a sensor reading at node i change?"

Node feature vector (uncertain):

Edge Perturbation (ε)



Each edge's features can vary:

$$e_{ij} \in [\bar{e}_{ij} - \varepsilon, \bar{e}_{ij} + \varepsilon]$$

"How much can the relationship between nodes i and j change?"

Edge feature (uncertain):

Key Takeaways

- GraphStar sets extend Star sets from vectors to $N \times F$ matrices — preserving graph structure
- Perturbation in GNNs spreads across topology, not just features
- Verification: $\hat{A} \cdot V \cdot W + b$ (linear, constraints unchanged) $\rightarrow$ flatten $\rightarrow$ ReLU reach. $\rightarrow$ reshape
- Subgraph verification exploits k -hop locality for **exact**, scalable results
- Node and edge perturbations are independent $\rightarrow$ block-diagonal constraints $\rightarrow$ tractable